Total No. of Questions : 8]

SEAT No. :

PC2798

[Total No. of Pages : 3

[6352]-22

S.E. (E & TC)

CONTROL SYSTEMS

(2019 Pattern) (Semester - IV) (204192)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Figures to the right indicate full marks.*
3) *Assume suitable data, if necessary.*

Q1) a) Investigate the stability of the system using Routh Hurwitz criterion **[8]**

$$G(S)=\frac{100}{S^4+6S^3+30S^2+60S+100}$$

b) The O.L.T.F of unity gain negative feedback system given **[10]**

$$G(S)=\frac{k}{s(s+4)(s^2+s+1)}$$

i) Calculate the range of k for system to be in stable state when stability of closed loop system is concerned

ii) Calculate the value of k for system to become marginally stable, also calculate the frequency of natural oscillation

OR

Q2) a) The unity feedback system has open loop transfer function $G(S)=\frac{k}{s(s+2)(s+5)(s+10)}$. Determine the range of 'k' for the system stability, value of 'k' and frequency of oscillation at marginal stability. **[8]**

b) A unity feedback transfer function has forward path gain $G(S)=\frac{k}{s(s+2)}$ Plot a root locus. **[10]**

P.T.O.

Q3) a) If $G(S)H(S) = \frac{24}{S(S+2)(S+12)}$, Construct the Bode plot and Calculate gain Crossover frequency, Phase Crossover frequency. **[9]**

b) Draw the Polar plot for given transfer function. $G(S) = \frac{10}{S+2}$ **[8]**

OR

Q4) a) For unity feedback system with open loop transfer function $G(S) = \frac{100}{S(S+9)}$. Determine damping factor, undamped natural frequency, resonant peak, and resonant frequency. **[9]**

b) Define and explain **[8]**

i) Bandwidth

ii) Gain margin

iii) Phase margin

iv) Gain cross-over frequency

v) Phase cross over frequency

Q5) a) A feedback system with transfer function $G(s) = \frac{S^2 + 3S + 3}{S^3 + 2S^2 + 3S + 1}$. Construct a state model for the system. **[9]**

b) Find Controllability and Observability of the system given by state model. **[9]**

$$A = \begin{pmatrix} -2 & 1 & 0 \\ 1 & -3 & 2 \\ 10 & 0 & -8 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0.1 \\ 1 \end{pmatrix} \quad C = [1\ 0\ 1] \quad D = [0]$$

OR

Q6) a) Explain advantages and disadvantages of Conventional Control Theory. **[9]**

b) Determine the State transition matrix of state equation **[9]**

$$x(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Q7) a) What do you mean by On-Off control? Explain with suitable example. **[9]**

b) Write comparison of P,I and D control action. **[8]**

OR

Q8) a) How IoT helps in Industrial Automation? What are the essentials of an Industrial IoT solution? Give two examples of Industrial IoT. **[9]**

b) Draw and explain the block diagram of digital control system. **[8]**

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